

CCNA

COMPLETE GLOSSARY

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Covering Networking Fundamentals, Ethernet, IPv4 & IPv6, Subnetting, Routing & Switching, VLANs & Trunking, Spanning Tree, EtherChannel, OSPF, NAT/PAT, DHCP, ACLs, FHRPs, Wireless, Security, Network Services, Automation, QoS, and more

A

Access Control Entry

Abbreviation: ACE

A single rule inside an access control list that permits or denies specific traffic.

Access Control List

Abbreviation: ACL

A set of rules used to permit or deny traffic based on criteria such as IP address or port.

Access Point

Abbreviation: AP

A device that lets wireless clients connect to a wired network.

Access Port

A switch port that carries traffic for only one VLAN, normally used to connect end devices.

Active Router

In HSRP, the router currently forwarding traffic sent to the virtual IP address.

Address Resolution Protocol

Abbreviation: ARP

A protocol that maps an IP address to a MAC address on a local network.

Administrative Distance

Abbreviation: AD

A number that shows how trustworthy a routing source is; lower values are preferred.

Ansible

An open-source automation tool used to configure devices and push changes without manual login.

Anycast Address

An IPv6 address type assigned to multiple devices, where traffic is delivered to the nearest one.

APIPA

Abbreviation: APIPA

A self-assigned IP address (169.254.x.x) a device uses when it cannot reach a DHCP server.

Application Programming Interface

Abbreviation: API

A defined way for software programs to talk to each other and exchange data.

Area

A logical grouping of routers in OSPF that share the same link-state database.

Authentication, Authorization, and Accounting

Abbreviation: AAA

A framework that controls who can access a network, what they can do, and logs what they did.

Auto-MDIX

A feature that lets a switch port automatically detect and adjust to the correct cable type.

Autonomous Access Point

A standalone wireless access point that has its own configuration and doesn't need a controller.

B

Bandwidth

The maximum amount of data that can theoretically travel through a link in a given time.

Bit

The smallest unit of data in networking, represented as a 0 or a 1.

Blocking State

An STP port state where the port does not forward frames, used to prevent loops.

Boot System

A Cisco IOS configuration command that tells a device which image file to load on startup.

Border Gateway Protocol

Abbreviation: BGP

A routing protocol used to exchange routes between different organizations on the internet.

BPDU Guard

A switch security feature that shuts down a port if it receives a BPDU, protecting the STP topology.

Bridge ID

Abbreviation: BID

A value made of a priority and a MAC address, used by STP to elect the root bridge.

Bridge Protocol Data Unit

Abbreviation: BPDU

A message switches exchange to build and maintain the Spanning Tree topology.

Broadcast Address

The last address in a subnet, used to send data to every device in that subnet.

Broadcast Domain

A group of devices that all receive each other's broadcast traffic.

C

Carrier Sense Multiple Access with Collision Detection

Abbreviation: CSMA/CD

A method Ethernet devices once used to detect and recover from collisions on a shared medium.

Cisco Discovery Protocol

Abbreviation: CDP

A Cisco proprietary protocol that lets directly connected devices learn about each other.

Cisco IOS

Abbreviation: IOS

The operating system that runs on most Cisco routers and switches.

Class of Service

Abbreviation: CoS

A QoS marking field in an Ethernet frame used to prioritize traffic at Layer 2.

Classful Addressing

An older IP addressing method where the subnet mask is fixed based on the address class.

Classification and Marking

The QoS process of identifying traffic type and tagging it so it can be treated appropriately.

Classless Addressing

An IP addressing method where the subnet mask can be adjusted instead of following fixed classes.

Client-Server Model

A network design where clients request services and a server provides them.

Collision Domain

A network segment where data frames can collide with each other if sent at the same time.

Congestion

A network condition where more traffic arrives than a link or device can handle at once.

Console Port

A physical port used to connect directly to a device for initial configuration.

Convergence

The time it takes for all routers in a network to agree on the current topology after a change.

Crossover Cable

An Ethernet cable used to connect two similar devices, such as switch to switch.

Cut-Through Switching

A switching method that starts forwarding a frame as soon as the destination address is read.

D

Data Link Layer

Layer 2 of the OSI model, responsible for MAC addressing and delivering frames on a local network.

Dead Timer

In OSPF, the amount of time a router waits without hearing from a neighbor before declaring it down.

Decapsulation

The process of removing headers as data moves up through the layers of the OSI model.

Default Gateway

The router address a device uses to send traffic destined for another network.

Default Route

A route used to forward traffic when no specific matching route exists in the routing table.

Demilitarized Zone

Abbreviation: DMZ

A separate network segment that hosts public-facing servers while keeping the internal network protected.

Denial of Service

Abbreviation: DoS

An attack that tries to make a network resource unavailable by overwhelming it with traffic.

Designated Port

The STP port on a network segment that is chosen to forward traffic.

Designated Router

Abbreviation: DR

In OSPF, the router elected on a multi-access network to reduce the number of adjacencies formed.

Differentiated Services Code Point

Abbreviation: DSCP

A field in the IP header used to mark packets for QoS treatment.

Directly Connected Route

A route added automatically to the routing table for a network attached to a router's interface.

Distributed Denial of Service

Abbreviation: DDoS

A DoS attack launched from many different sources at the same time.

DNS Resolver

A service or component that looks up a domain name and returns its matching IP address.

Domain Name System

Abbreviation: DNS

A service that translates human-friendly domain names into IP addresses.

Dual Stack

A configuration where a device runs both IPv4 and IPv6 at the same time.

Dynamic Host Configuration Protocol

Abbreviation: DHCP

Automatically assigns IP addresses and other network settings to devices.

Dynamic NAT

A form of NAT that maps private addresses to a pool of public addresses on a first-come basis.

Dynamic Routing

A method where routers automatically learn and share routes using a routing protocol.

Dynamic Trunking Protocol

Abbreviation: DTP

A Cisco protocol that automatically negotiates trunking between two connected switches.

EtherChannel

A technology that bundles multiple physical links into one logical link for more bandwidth and redundancy.

Ethernet

The most common wired networking standard used to connect devices in a LAN.

Ethernet Frame

The formatted unit of data, including headers and trailer, that Ethernet uses to move data on a LAN.

EUI-64

A method used to automatically generate the interface portion of an IPv6 address from a MAC address.

Extended Access Control List

An ACL type that filters traffic based on source and destination address, protocol, and port.

Extended Service Set

Abbreviation: ESS

Two or more access points working together to provide continuous wireless coverage.

F

File Transfer Protocol

Abbreviation: FTP

A protocol used to transfer files between a client and a server over a network.

First Hop Redundancy Protocol

Abbreviation: FHRP

A group of protocols that provide backup default gateways so hosts stay connected if a router fails.

Flash Memory

Storage inside a router or switch used to hold the IOS image file.

Floating Static Route

A backup static route with a higher administrative distance, used only if the primary route fails.

Flow Control

A mechanism that prevents a sender from overwhelming a receiver with too much data at once.

Forwarding State

The STP port state where a port is fully active and sending and receiving frames.

Fragmentation

The process of breaking a packet into smaller pieces so it can fit within a link's maximum size.

Frame

A Layer 2 unit of data that includes source and destination MAC addresses.

Frame Check Sequence

Abbreviation: FCS

A field at the end of an Ethernet frame used to detect transmission errors.

Fully Qualified Domain Name

Abbreviation: FQDN

A complete domain name that specifies an exact location in the DNS hierarchy, like `www.example.com`.

G

Gateway Load Balancing Protocol

Abbreviation: GLBP

A Cisco FHRP that allows multiple routers to share the traffic load while providing gateway redundancy.

Gateway of Last Resort

The default route a router uses when no other matching route is found.

Generic Routing Encapsulation

Abbreviation: GRE

A tunneling protocol that encapsulates a wide variety of network layer protocols inside virtual point-to-point links.

Gratuitous ARP

An ARP message a device sends without being asked, usually to announce or update its own IP-to-MAC mapping.

H

Half Duplex

A communication mode where a device can send or receive data, but not both at the same time.

Hello Packet

A message routers send to discover and maintain relationships with neighboring routers.

Host Address

An IP address assigned to a specific device within a subnet.

Hot Standby Router Protocol

Abbreviation: HSRP

A Cisco FHRP that lets a group of routers share a virtual IP address for gateway redundancy.

HTTP Secure

Abbreviation: HTTPS

A secure, encrypted version of HTTP used to protect web traffic.

Hub

An older Layer 1 device that repeats incoming signals out all other ports, creating one collision domain.

Hypertext Transfer Protocol

Abbreviation: HTTP

The protocol used to transfer web pages between a browser and a web server.



Implicit Deny

The rule at the end of every ACL that blocks any traffic not specifically permitted.

Inside Global Address

The public IP address that represents an internal host to the outside network after NAT.

Inside Local Address

The private IP address of a host on the inside network before NAT translation.

Inter-VLAN Routing

The process of routing traffic between different VLANs, usually done by a router or Layer 3 switch.

Internet Control Message Protocol

Abbreviation: ICMP

A protocol used to send error messages and diagnostic information, such as with ping.

Intrusion Detection System

Abbreviation: IDS

A security tool that monitors traffic and alerts on suspicious activity without blocking it.

Intrusion Prevention System

Abbreviation: IPS

A security tool that monitors traffic and can actively block detected threats.

IPv4

The 32-bit addressing system that has traditionally been used to identify devices on a network.

IPv6

The 128-bit addressing system designed to replace IPv4 and provide a much larger address space.

J

Jitter

The variation in delay between packets arriving, which can affect voice and video quality.

JSON

Abbreviation: JSON

A lightweight text format used to structure data, commonly used by APIs and automation tools.

L

Latency

The amount of time it takes for data to travel from source to destination.

Layer 3 Switch

A switch that can perform routing functions in addition to normal Layer 2 switching.

Lease Time

The length of time a DHCP client is allowed to use an assigned IP address.

Lightweight Access Point

A wireless access point that depends on a wireless LAN controller for its configuration.

Link Aggregation

The general concept of combining multiple physical links into a single logical connection.

Link Aggregation Control Protocol

Abbreviation: LACP

An open standard protocol used to negotiate and manage EtherChannel links between two devices.

Link-Local Address

An IPv6 address automatically used for communication only within the local network segment.

Link-State Advertisement

Abbreviation: LSA

A message OSPF routers exchange to describe their links and neighbors.

Link-State Database

Abbreviation: LSDB

A table each OSPF router builds containing information about the entire network topology.

Local Area Network

Abbreviation: LAN

A network that connects devices within a small geographic area, such as a home or office.

Longest Match Rule

The rule a router uses to pick the most specific matching route when several routes could apply.

Loopback Address

A special address used by a device to test its own network stack, such as 127.0.0.1.

Loopback Interface

A virtual interface on a router that is always up and often used for stable management or routing IDs.

M

MAC Address

A unique hardware address burned into a network interface, used for Layer 2 communication.

MAC Address Table

A table a switch keeps that maps MAC addresses to the ports they were learned on.

MAC Flooding

An attack that overwhelms a switch's MAC address table to force it to flood traffic to all ports.

Malware

Malicious software designed to damage, disrupt, or gain unauthorized access to a system.

Management Information Base

Abbreviation: MIB

A database structure that defines what information can be monitored or managed through SNMP.

Management VLAN

A VLAN dedicated to managing network devices, kept separate from regular user traffic.

Maximum Transmission Unit

Abbreviation: MTU

The largest size a packet is allowed to be on a given network link.

Metric

A value routing protocols use to measure and compare the cost of different paths.

Metropolitan Area Network

Abbreviation: MAN

A network that spans a city or a large campus, bigger than a LAN but smaller than a WAN.

Multicast Address

An address used to deliver traffic to a specific group of interested devices at once.

N

Native VLAN

The VLAN on a trunk link whose traffic is sent untagged.

Neighbor

In routing protocols, another router that a device has formed a direct relationship with to exchange routes.

Network Address

The address that identifies a subnet as a whole, always the first address in the range.

Network Address Translation

Abbreviation: NAT

A method that translates private IP addresses to public ones so devices can reach the internet.

Network Automation

The practice of using software to configure, manage, and monitor networks instead of doing it manually.

Network Interface Card

Abbreviation: NIC

The hardware component that connects a device to a network.

Network Time Protocol

Abbreviation: NTP

A protocol used to synchronize the clocks of devices across a network.

Next-Hop Address

The IP address of the next router a packet should be sent to on its way to the destination.

NVRAM

Non-volatile memory in a router or switch that stores the startup configuration even after a reboot.

O

Open Shortest Path First

Abbreviation: OSPF

A link-state routing protocol that routers use to dynamically learn and share network paths.

OSI Model

A seven-layer reference model that describes how data moves through a network.

Outside Global Address

The public IP address of a destination host as seen from outside the local network.

Outside Local Address

The address of an outside destination as it appears to inside hosts, before any translation.

P

Password Authentication

A basic security method that requires a correct password before granting access to a device.

Peer-to-Peer Network

A network design where devices share resources directly with each other without a dedicated server.

Phishing

A social engineering attack that tricks users into revealing sensitive information through fake messages or sites.

Ping

A tool that uses ICMP to test whether a remote device is reachable on the network.

Port Address Translation

Abbreviation: PAT

A form of NAT that lets many private addresses share a single public address using different port numbers.

Port Aggregation Protocol

Abbreviation: PAgP

A Cisco proprietary protocol used to negotiate and manage EtherChannel links.

Port Cost

A value STP assigns to a port based on link speed, used to calculate the best path to the root bridge.

Port Mirroring

Abbreviation: SPAN

A feature that copies traffic from one switch port to another so it can be monitored or analyzed.

Port Security

A switch feature that restricts which devices can connect to a port based on MAC address.

Power-On Self-Test

Abbreviation: POST

A hardware check a device runs on startup before loading its operating system.

Preamble

A pattern at the start of an Ethernet frame that helps devices synchronize before the actual data arrives.

Prefix Length

The number of bits in an IP address that represent the network portion, written after a slash.

Private IP Address

An IP address reserved for use inside local networks and not routable on the public internet.

Protocol

A set of rules that defines how devices communicate and exchange data.

Protocol Data Unit

Abbreviation: PDU

The general term for a unit of data at any layer of the OSI model, such as a frame or packet.

Public IP Address

A unique IP address that is routable on the internet.

Q

Quality of Service

Abbreviation: QoS

A set of techniques used to manage bandwidth and prioritize certain types of traffic over others.

Queuing

The process of holding packets temporarily so they can be sent in a prioritized order.

R

Rapid Spanning Tree Protocol

Abbreviation: RSTP

An updated version of Spanning Tree Protocol that converges much faster after a topology change.

Read-Only Memory

Abbreviation: ROM

Memory in a device that stores basic startup instructions and normally cannot be changed.

Recursive Route Lookup

The process a router uses to keep searching its routing table until it finds a directly connected next hop.

Remote Authentication Dial-In User Service

Abbreviation: RADIUS

A protocol used to centralize authentication, authorization, and accounting for network access.

Repeater

A device that regenerates a signal to extend the distance it can travel on a network.

REST API

A common style of API that uses standard web methods to let systems exchange data over HTTP.

Reverse Address Resolution Protocol

Abbreviation: RARP

An older protocol that let a device learn its own IP address using its known MAC address.

Rogue DHCP Server

An unauthorized DHCP server on a network that can hand out incorrect or malicious settings.

Rollover Cable

A cable used to connect a computer to a device's console port for direct configuration.

ROMMON

A minimal bootstrap environment used to recover or reload the IOS on a Cisco device.

Root Bridge

The switch elected as the reference point for the Spanning Tree topology.

Root Port

The port on a non-root switch that provides the best path back to the root bridge.

Route Summarization

The practice of combining multiple smaller routes into one larger route to reduce routing table size.

Routed Protocol

A protocol, such as IPv4 or IPv6, that carries user data and can be routed between networks.

Router

A device that forwards traffic between different networks based on IP addresses.

Router ID

A unique identifier, often an IP address, that identifies a router within a routing protocol like OSPF.

Router-on-a-Stick

A design where a single router interface, using subinterfaces, handles routing for multiple VLANs.

Routing Information Protocol

Abbreviation: RIP

A simple distance-vector routing protocol that uses hop count to choose the best path.

Routing Protocol

A protocol routers use to share information about networks and build their routing tables.

Routing Table

A list a router keeps of known networks and how to reach them.

S

Secure Shell

Abbreviation: SSH

A secure protocol used to remotely manage devices with encrypted communication.

Segment

A Layer 4 unit of data, typically associated with TCP.

Service Set Identifier

Abbreviation: SSID

The name broadcast by a wireless network so users can find and connect to it.

Severity Level

A ranking used in Syslog messages that shows how serious an event is.

Simple Mail Transfer Protocol

Abbreviation: SMTP

A protocol used to send email between mail servers.

Simple Network Management Protocol

Abbreviation: SNMP

A protocol used to monitor and manage devices on a network from a central system.

SNMP Agent

Software running on a managed device that collects and reports information to an SNMP manager.

SNMP Manager

The central system that collects and processes information from SNMP agents across the network.

SNMP Trap

An unsolicited alert an SNMP agent sends to notify the manager of a significant event.

Social Engineering

An attack technique that manipulates people into giving up confidential information or access.

Socket

The combination of an IP address and port number that identifies a specific communication session.

Software-Defined Networking

Abbreviation: SDN

An approach that separates network control from the hardware, allowing centralized management through software.

Spanning Tree Protocol

Abbreviation: STP

A protocol that prevents Layer 2 loops by blocking redundant paths between switches.

Spoofing

An attack where a device disguises itself as another device, such as by faking its IP or MAC address.

Standard Access Control List

An ACL type that filters traffic based only on the source IP address.

Standby Router

In HSRP, the router ready to take over forwarding duties if the active router fails.

Static NAT

A form of NAT that maps one private address to one public address on a permanent basis.

Static Route

A route that an administrator manually configures rather than one learned dynamically.

Store-and-Forward Switching

A switching method that receives the entire frame and checks it for errors before forwarding it.

Straight-Through Cable

An Ethernet cable used to connect two different types of devices, such as a PC to a switch.

Stratum

A number in NTP that shows how many hops a device is from an authoritative time source.

Subinterface

A logical division of a physical interface, often used to handle traffic for a specific VLAN.

Subnet

A smaller network created by dividing a larger network into pieces.

Subnet Mask

A number that identifies which portion of an IP address represents the network and which represents the host.

Subnetting

The process of dividing a larger network into smaller, more manageable subnetworks.

Supernetting

The process of combining multiple smaller networks into one larger network.

Switch

A Layer 2 device that forwards frames between devices based on MAC addresses.

Switch Virtual Interface

Abbreviation: SVI

A virtual Layer 3 interface on a switch that allows routing for a specific VLAN.

Switching

The process of forwarding frames within the same network based on MAC address information.

Syslog

A standard used to send log messages from devices to a central server for storage and review.

TCP/IP Model

A four-layer networking model that describes how data is prepared and sent across a network.

Telnet

An older, unencrypted protocol used to remotely access and manage devices.

Three-Way Handshake

The SYN, SYN-ACK, ACK process TCP uses to establish a reliable connection.

Throughput

The actual amount of data successfully transferred over a link in a given time.

Time to Live

Abbreviation: TTL

A value in a packet that limits how many hops it can take before being discarded.

Topology

The physical or logical layout of how devices are connected in a network.

Topology Change Notification

Abbreviation: TCN

A message switches use in STP to inform others that the network topology has changed.

Traceroute

A diagnostic tool that shows the path packets take across multiple routers to reach a destination.

Traffic Policing

A QoS technique that drops or marks down traffic that exceeds a defined rate.

Traffic Shaping

A QoS technique that delays excess traffic to smooth it out instead of dropping it.

Transmission Control Protocol

Abbreviation: TCP

A connection-oriented Layer 4 protocol that guarantees reliable, ordered delivery of data.

Trivial File Transfer Protocol

Abbreviation: TFTP

A simple, lightweight file transfer protocol often used to move IOS images or configuration files.

Trunk Port

A switch port configured to carry traffic for multiple VLANs, usually between switches.

U

Unicast Address

An address that identifies a single, specific device on a network.

User Datagram Protocol

Abbreviation: UDP

A connectionless Layer 4 protocol that sends data quickly without guaranteeing delivery.

User EXEC Mode

The most basic Cisco IOS command mode, allowing only limited, non-configuration commands.

V

Virtual IP Address

Abbreviation: VIP

A shared IP address used by an FHRP group so hosts always have a consistent default gateway.

Virtual LAN

Abbreviation: VLAN

A logical grouping of devices on a switch that behaves as its own separate network.

Virtual Private Network

Abbreviation: VPN

A technology that creates a secure, encrypted connection over a public network.

Virtual Router Redundancy Protocol

Abbreviation: VRRP

An open standard FHRP that provides gateway redundancy similar to HSRP.

Virtual Terminal Line

Abbreviation: VTY

A logical line on a Cisco device used for remote access sessions such as Telnet or SSH.

VLAN Hopping

An attack technique that lets traffic move between VLANs it shouldn't normally have access to.

VLAN Trunking Protocol

Abbreviation: VTP

A Cisco protocol that lets switches share VLAN configuration information automatically.

Voice VLAN

A VLAN dedicated to carrying voice traffic, often prioritized for better call quality.

W

Wi-Fi Protected Access 2

Abbreviation: WPA2

A wireless security standard that uses strong encryption to protect Wi-Fi networks.

Wi-Fi Protected Access 3

Abbreviation: WPA3

The newer wireless security standard offering stronger protections than WPA2.

Wide Area Network

Abbreviation: WAN

A network that connects devices across large distances, often between cities or countries.

Wildcard Mask

A mask used with OSPF and ACLs that works opposite to a subnet mask to define a range of addresses.

Wireless LAN Controller

Abbreviation: WLC

A central device that manages and configures multiple lightweight access points.

Y

YANG

Abbreviation: YANG

A data modeling language used to define the structure of configuration and state data for network automation.

Z

Zero Trust

A security approach that assumes no user or device should be trusted by default, even inside the network.
